Question:

Carlos is planting a lemon tree. The tree will cost $90 to plant. Each year it will grow 7 lemons, which he can sell for $1.5 each. It costs $3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

<tool_call> Reasoning process here:

Step 1: Understand the setup.
- Initial cost to plant the tree: $90 (one-time cost).
- Each year, the tree produces 7 lemons.
- Each lemon sells for $1.50.
- Annual cost to water and feed: $3.
- We need to find the number of years until Carlos starts earning money, i.e., when the income from lemons exceeds the annual expenses (including the initial cost being amortized over time).

Step 2: Determine annual profit.
Each year:
- Revenue from lemons = 7 lemons ÃĹ $1.50 = $10.50.
- Cost to water and feed = $3.
- So, annual profit = $10.50 - $3 = $7.50.

But this is only after the initial $90 cost. We need to determine when the cumulative profit from the tree (after covering the initial cost) becomes positive, i.e., when the total income exceeds the total cost.

Let âĢĻs calculate the total cost after n years:
- Initial cost: $90
- Annual cost (water & feed): $3 per year âĨĴ after n years: 3n
- Total cost after n years = 90 + 3n

Total income from lemons after n years:
- Each year: 7 ÃĹ 1.5 = $10.50 âĨĴ total income = 10.5n

We need to find the smallest integer n such that:
Total income > Total cost
10.5n > 90 + 3n

Step 3: Solve the inequality
10.5n > 90 + 3n
10.5n - 3n > 90
7.5n > 90
n > 90/7.5
n > 12

So, n > 12 âĨĴ the smallest integer n is 13.

Now verify:
After 12 years:
- Income = 10.5 ÃĹ 12 = $126
- Cost = 90 + 3 ÃĹ 12 = 90 + 36 = $126 âĨĴ break even, not earning money yet.

After 13 years:
- Income = 10.5 ÃĹ 13 = 136.5
- Cost = 90 + 3 ÃĹ 13 = 90 + 39 = 129
- Profit = 136.5 - 129 = $7.5 âĨĴ he starts earning money in year 13.

Therefore, he starts earning money in year 13.

< answer > 13 </ answer >